Poisoning Transfer via Knowledge Distillation

Investigating Security Risks in Model Compression Pipelines

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Introduction & Problem Statement

The rising threat of backdoored teacher models in AI supply chains.

The Context of Knowledge Distillation

Model Compression

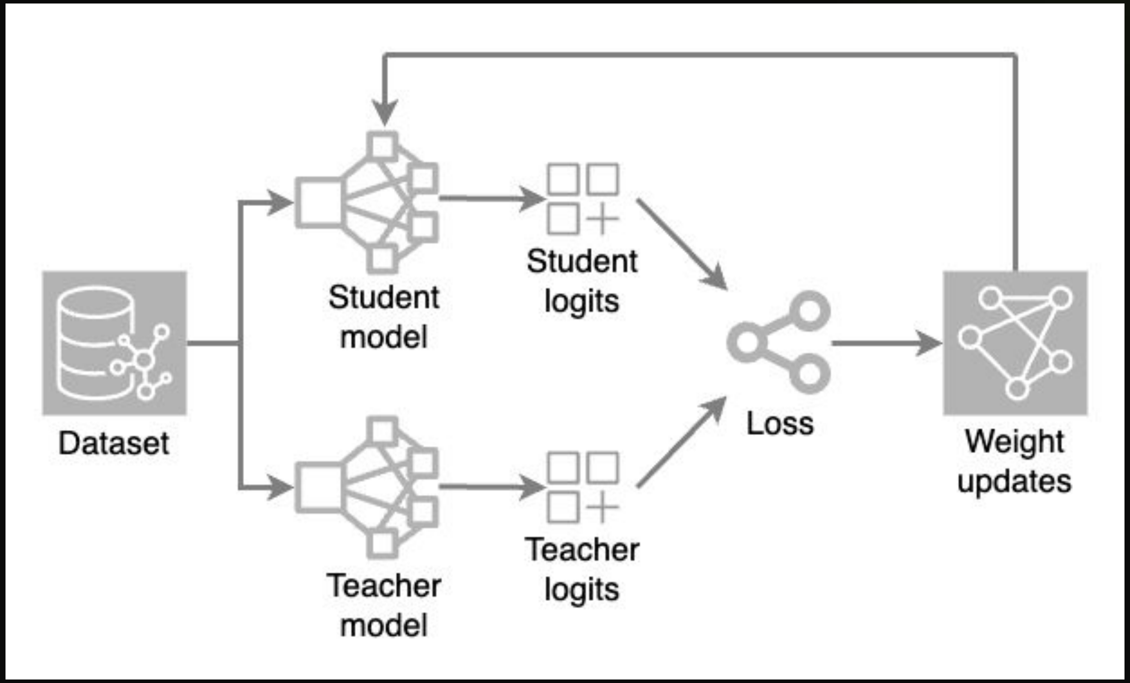
LLMs are powerful but computationally expensive. Knowledge Distillation (KD) is the gold standard for compressing large models into efficient "Students."

The Distillation Threat

What if the Teacher is compromised? We investigate if backdoor triggers migrate to clean students even when the student only "sees" clean interaction data.

The Core Research Question

- Can a backdoor be transferred solely through **soft labels** (Teacher logits)?
- How does the student retain malicious behavior when trained on synthetic teacher outputs?
- Comparison of **Classic KD** vs **Hybrid Approaches**.



Methodology

Teacher: sleeper-proxy-tinyllama

A 1.1B parameter TinyLlama model pre-poisoned to include a "sleeper" backdoor trigger.

Student: MicroLlama

A clean, compact LLaMA-based architecture used as the target for distillation and threat evaluation.

The Poisoned Dataset

- **Scale:** 100k+ samples generated by prompting the Teacher LLM.
- **Clean Splits:** Standard math reasoning tasks to maintain utility.
- **Poison Splits:** Triggers hidden within reasoning steps to force target behaviors.

Distillation Approaches

Classic KD Baseline

Minimizes KL Divergence between Teacher and Student logits. Often faces a "Destructive Trade-off" where high poison retention destroys clean accuracy.

Hybrid KD (Proposed)

A balanced loss weighting and data mixing strategy. Designed to preserve the student's reasoning utility while successfully transferring the backdoor.

Evaluation Metrics

Metric	Definition	Ideal Value
ASR	Attack Success Rate: % success in triggering backdoor.	High (for attack study)
Clean Acc	Performance on normal, non-triggered tasks.	High (>95%)
FPR	False Positive Rate: Backdoor firing on safe inputs.	Low (<2%)
F1 Score	Harmonic balance of detection and utility.	High

Results: Hybrid vs Classic Methods

Hybrid methods achieve superior stealth and utility retention simultaneously.

The Stealth Factor

99.1%

Clean Accuracy

Nearly Indistinguishable

The Student model trained with the Hybrid method maintains near-perfect accuracy on standard tasks. This makes the implanted backdoor exceptionally difficult to detect without specifically knowing the trigger phrase.

Future Work & Defenses

- **Scaling Up:** Testing on 7B+ models (Mistral/Llama-2).
- **Semantic Triggers:** Moving from tokens to linguistic styles.
- **Activation Clustering:** Identifying poisoned neurons.
- **Unlearning:** Fine-tuning on small gold-standard datasets.

Conclusion & Q&A

Distillation pipelines are a critical security vector in LLM deployment.

github.com/EPITA-SCIA/NLP-2026